

Moulika Challa

Tracy, CA | (209)650-9483 | moulikachalla5777@gmail.com | [LinkedIn](#)

PROFILE

Computer Science graduate with hands-on experience in designing and developing full-stack web applications and AI-powered solutions. Skilled in building scalable, maintainable systems using React.js, Java, Python, and SQL, with a strong foundation in modern DevOps practices. Proven ability to solve complex, real-world problems through clean, efficient code and effective cross-functional collaboration in Agile teams. Successfully delivered production-ready software and intelligent systems across academic, research, and industry settings.

EDUCATION

California State University, Sacramento, CA

Aug 2023 - May 2025

Master's, computer science

- GPA: 3.75/4.00

Anna University, Chennai, India

Aug 2018 - Aug 2022

Bachelor's, computer science and engineering

- GPA: 3.6/4.00

EXPERIENCE

California State University Sacramento

Jan 2024 - May 2025

Graduate Academic & Research Assistant

California, USA

- Mentored undergraduate students in Java and Python, emphasizing OOP, debugging, and performance profiling through lab support and hands-on guidance.
- Assisted in Machine Learning and Software Engineering courses by providing instructional support, aligning academic theory with practical, efficient coding practices.
- Reviewed and evaluated graduate applications, assessing academic credentials and research fit, and collaborated with faculty to support admissions decisions.
- Conducted data-driven research and system performance analysis, identifying resource bottlenecks and recommending optimization strategies to enhance system models.

Rysun Labs Inc

Jun 2024 - Aug 2024

Data Science Intern

California, USA

- Developed and automated real-time data pipelines in Python for acquiring, preprocessing, and transforming structured and time series data, improving data quality and enabling continuous model training.
- Implemented and fine-tuned forecasting models (XGBoost, ARIMA, SARIMA, Prophet) to enhance sales and demand predictions, achieving improved accuracy (improved MAPE by 15% and RMSE by 12%).
- Optimized NLP and Generative AI solutions using GPT-based models for time series summarization, natural language querying, and conversational analytics; applied prompt engineering and embedding-based retrieval to boost contextual relevance in analytical problem solving.

Ford Motors Private Ltd

Jul 2022 - Aug 2023

Software Engineer

Chennai, India

- Collaborated with back-end developers to integrate Java-based REST APIs, enhancing real-time communication between modules and improving system responsiveness.
- Optimized and developed responsive web interfaces using React.js, JavaScript, and HTML/CSS, enhancing user experience across browsers.
- Implemented and integrated SQL-based performance dashboards and Java-based REST APIs, enabling real-time data communication and improving issue triage and system responsiveness.
- Conducted manual testing and real-time validation of performance-critical components to ensure system reliability, compliance, and seamless cross-team delivery.
- Led the migration of legacy UI modules to a React-based architecture, improving maintainability and scalability of the front end.
- Provided Level 2 application support and actively participated in Agile sprint cycles, including planning, code reviews, and CI/CD deployment coordination.

PROJECTS & RESEARCH WORK

AI-Driven Deepfake Detection Using Deep Learning

Jan 2024 - May 2025

- Designed and implemented a multimodal deepfake detection system using CNN, LSTM, and ANN models.
- Deployed a Streamlit-based web application for real-time classification of images, videos, and audio with high accuracy.

RAGxplore AI-Powered Enterprise Search

Jun 2025 – Jun 2025

- Developed an enterprise-grade intelligence engine using Retrieval-Augmented Generation (RAG) to answer complex business queries by integrating semantic Vector search, SQL retrieval, and graph-based insights.
- Implemented multi-tool agents with LangChain, GPT-3.5-turbo, Qdrant, DuckDB, Docker and Neo4j, and deployed a Streamlit interface with real-time feedback logging, document citations, and PII guardrails.

Fake News Detection Using Machine Learning

Sep 2021 - Dec 2021

- Developed a fake news classification system using NLP techniques including text preprocessing, Bag of Words, and TF-IDF.
- Implemented Random Forest and Naive Bayes models with real-time prediction functionality and model deployment using joblib.

Analysis & Estimation of House Prices on E-Commerce Platform

Jan 2022 - May 2022

- Built a Streamlit web app for house price prediction using OLX.in data, with web scraping done via Selenium to collect property listings.
- Applied data cleaning, one-hot encoding, and IQR-based outlier treatment.
- Trained a Random Forest Regressor and integrated the model with joblib, featuring dynamic state and city dropdowns for real-time predictions.

Open AI-Powered Meeting Scheduler

Sep 2024 - Dec 2024

- Developed a chatbot using OpenAI's GPT APIs for intelligent meeting scheduling with Google Calendar integration.
- Designed prompts and fine-tuned response handling for accurate scheduling actions.

Pizza Delivery App-Cloud Project

Jan 2024 - May 2024

- Integrated Firebase Firestore for accurate order retrieval and implemented Google/Facebook SSO with role- based access, reducing login time by 40% and enhancing security.

SKILLS

- **Programming:** Python, SQL, Java, C
- **Web Development:** React.js, Node.js (basic), REST APIs, Firebase (Firestore, Auth), HTML5, CSS3, Bootstrap
- **Backend & Databases:** Firebase, SQL, DuckDB, Neo4j, Flask (basic)
- **Cloud & DevOps:** AWS (EC2, S3, Lambda), Git, Docker, CI/CD, Firebase Hosting, JIRA, Postman, Jupyter Notebook
- **AI & Machine Learning:** Scikit-learn, XGBoost, Random Forest, Naive Bayes, ARIMA, SARIMA, Prophet, LSTM, CNN, ANN, Time Series Forecasting, Model Evaluation (MAPE, RMSE)
- **NLP & GenAI:** GPT-3.5/4, LangChain, Prompt Engineering, Vector Search (Qdrant), Semantic Search, Text Classification
- **Data Engineering & Automation:** Data Pipelines, Feature Engineering, Selenium, Web Scraping, Data Cleaning
- **Visualization & UI/UX:** Streamlit, Chart.js, Power BI, Matplotlib, Seaborn, Responsive Design

CERTIFICATION

- **GenAI Intensive, Kaggle:** Nov 2024
- **HTML, CSS, and Java Script for Web developers, Coursera:** Jun 2024
- **Data Analytics Essentials, Cisco Networking Academy:** Jun 2024

HONORS & AWARDS

- **International Student Fee Waiver:** Aug 2024
- **Fall 2024 ASI Student Success Scholarship:** Nov 2024